

KKBox Churn & Revenue Risk Analytics

Business Report

Live Tableau dashboard: https://public.tableau.com/app/profile/tianyi.ren4423/viz/Book1_17777346522100/D1_Overview

Executive Summary

This project turns KKBox subscription data into a practical retention analytics workflow. It predicts churn risk, quantifies revenue exposure, and translates model outputs into targeted retention actions that can be tested by a business team.

The project covers the full analytics lifecycle:

1. Build a local DuckDB analytical warehouse.
2. Transform raw member, transaction, listening, and churn-label data with dbt.
3. Create business-ready marts for KPIs, churn features, model outputs, segment risk, and monitoring.
4. Run statistical tests and churn modelling in Python.
5. Score users by churn probability and assign risk bands.
6. Quantify revenue at risk and retention ROI sensitivity.
7. Deliver Tableau dashboards and a final business report for stakeholder review.

Area	Result
Scored users	970,960
Observed churn rate	9.0%
Calibrated LightGBM AUC	0.805
Top-decile churn recall	39.7%
Top-decile lift	3.97x
Risk-adjusted revenue exposure	4.64M price units
Critical + high-risk concentration	1.47% of users / 71.5% of risk exposure

The main business finding is concentration. Critical and high-risk users are only a small share of the scored base, but they account for most of the modelled revenue exposure. Retention budget should therefore be targeted and tested, not spread evenly across all users.

End-to-End Architecture

End-to-End Analytics Architecture

From raw subscription data to governed marts, predictive scoring, dashboard communication, and retention decisions.

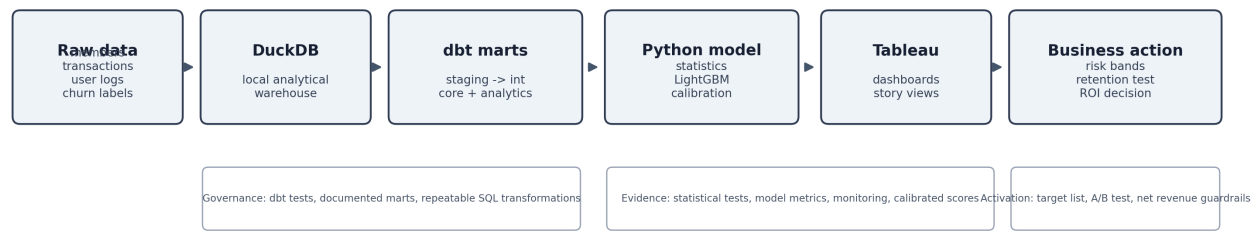


Figure 1: End-to-end analytics architecture

The workflow moves from raw source data to DuckDB, dbt marts, Python modelling, Tableau dashboards, and business actions. This mirrors an enterprise analytics process: raw data is transformed into trusted tables before it becomes a model, dashboard, or decision.

Data Foundation and dbt Lineage

Layer	What it does	Examples
Staging	Clean names, cast types, standardise raw columns	<code>stg_members</code> , <code>stg_transactions</code> , <code>stg_user_logs</code> , <code>stg_churn_labels</code>
Intermediate	Create reusable user-level summaries	member profile, transaction summary, engagement windows
Core marts	Build facts and dimensions	<code>dim_user</code> , subscription transactions, listening daily, churn labels
Analytics marts	Feed modelling and dashboards	churn features, segment summary, monthly KPIs, risk scores



Figure 2: dbt lineage graph

The lineage graph shows that final dashboards and model outputs are traceable back to source data and SQL transformation logic.

What Was Built

Analytical Warehouse

I created a DuckDB-based local warehouse that can process KKBBox source data without requiring a paid cloud setup. It supports user profile dimensions, subscription transaction facts, listening behaviour facts, churn labels, modelling features, dashboard KPIs, and user-level risk outputs.

dbt Transformation Layer

I used dbt to organise the SQL workflow into staging, intermediate, core, and analytics marts. This makes the project easier to rerun, document, and audit. In a business setting, this helps answer where each metric came from and whether dashboards and models are using consistent definitions.

Statistical Evidence

Test	Finding	Business use
Auto-renew off vs auto-renew on	67.8% churn vs 13.5% churn, $p < 0.001$	Auto-renewal status should be a core retention signal.
Discount band vs churn	$p < 0.001$	Discount policy is related to churn and should be evaluated financially.
Registration channel vs churn	$p < 0.001$, Cramer's V around 0.20	Acquisition channel can inform lifecycle targeting.

Churn Modelling

The modelling workflow compares an interpretable Logistic Regression baseline with a stronger LightGBM model. The final reporting model is calibrated LightGBM because it provides ranking power and usable probabilities for revenue-risk calculations.

Model	AUC	Log loss	Brier score	Top-decile recall	Top-decile lift
Logistic Regression	0.765	0.566	0.189	35.6%	3.56x
LightGBM raw	0.804	0.288	0.078	39.8%	3.98x
LightGBM calibrated	0.805	0.239	0.066	39.7%	3.97x

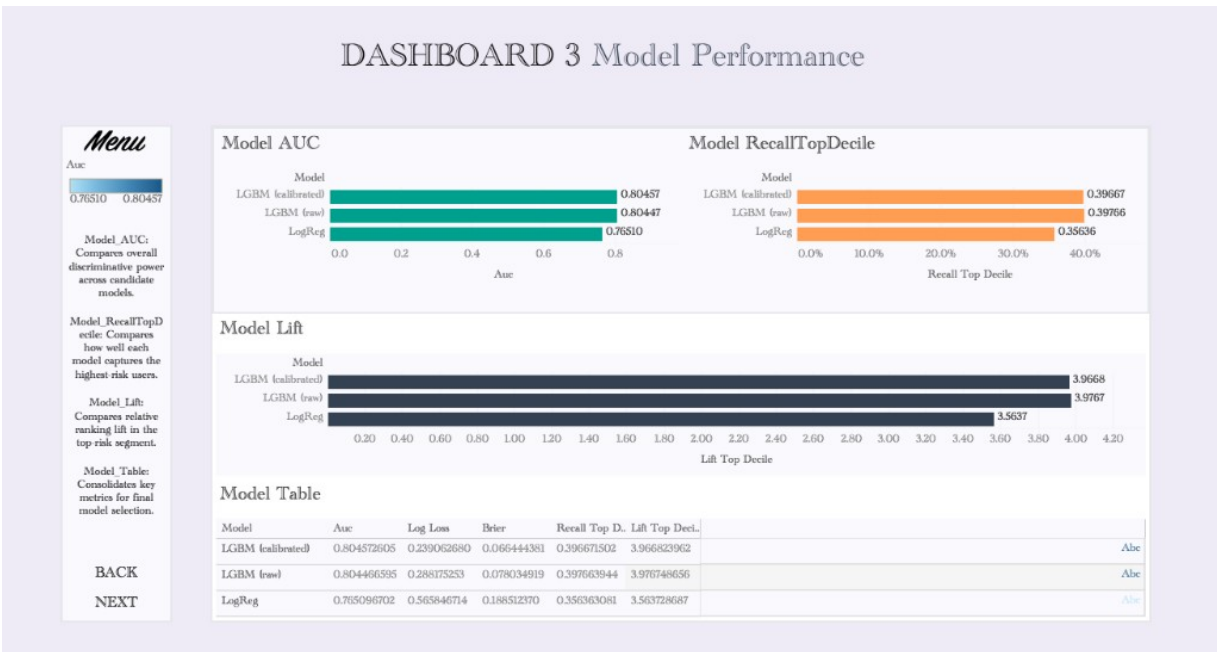


Figure 3: Model performance dashboard

Model Analysis Outputs from Jupyter

The Jupyter modelling notebook produces three main plots for model review: lift, calibration, and SHAP feature importance.

The lift chart shows whether the model is useful for operational targeting. At the top decile, the calibrated LightGBM model reaches about 3.97x lift, meaning it finds churners much more efficiently than random targeting.

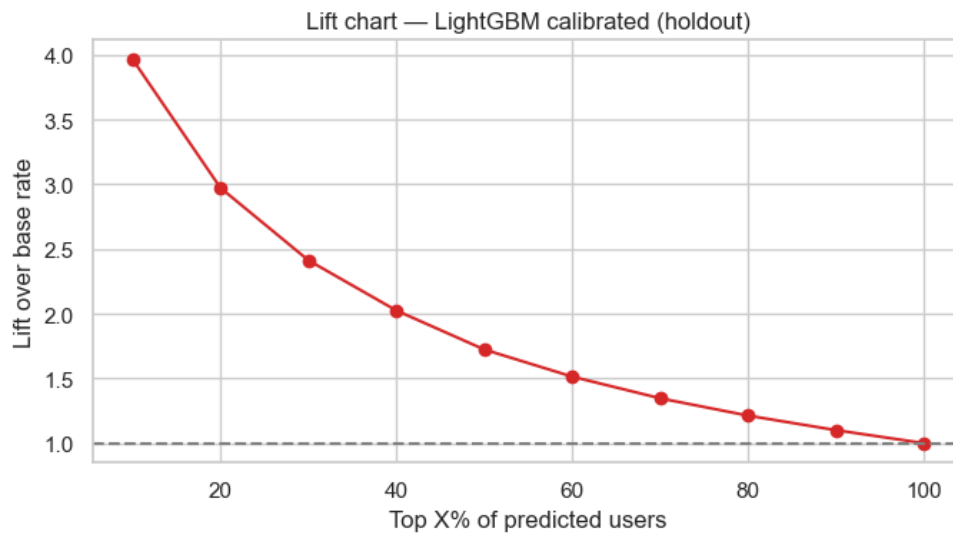


Figure 4: LightGBM calibrated lift chart

The calibration curve compares Logistic Regression, raw LightGBM, and calibrated LightGBM. Calibration matters because the project uses churn probabilities to estimate revenue at risk.

The SHAP importance plot explains which features have the largest average impact on the LightGBM churn prediction. This supports model interpretability and helps business stakeholders understand the drivers behind risk scores.

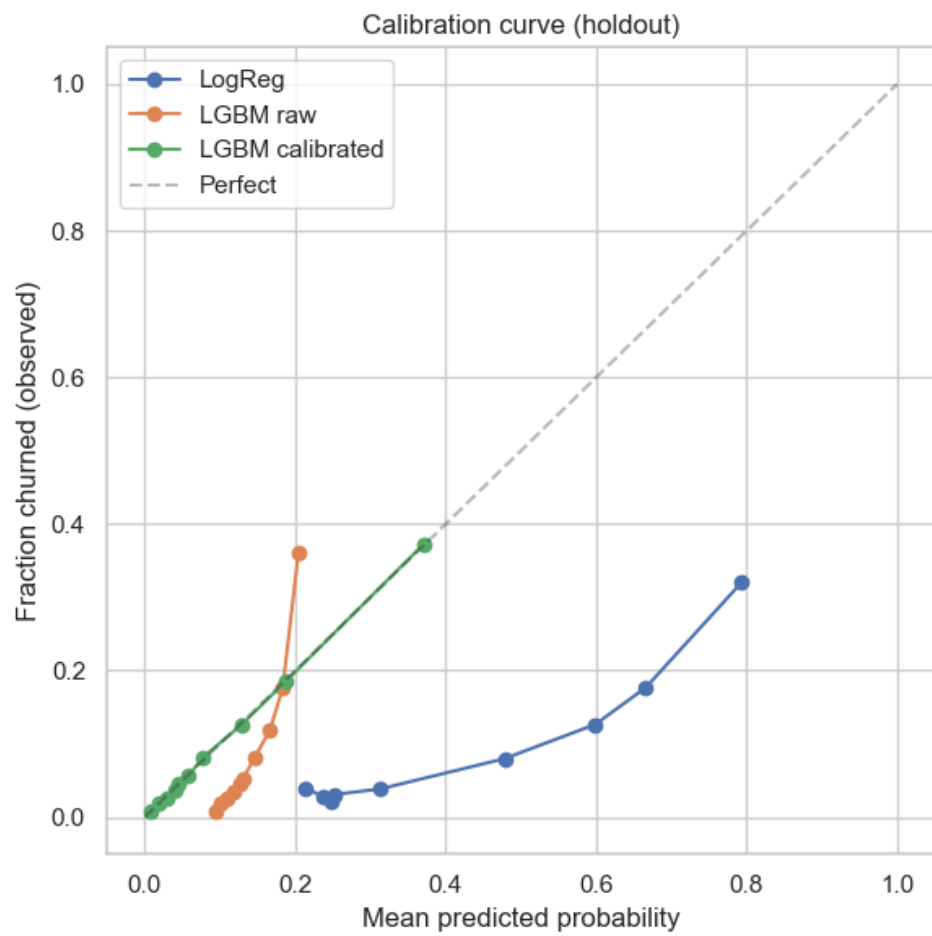


Figure 5: Model calibration curve

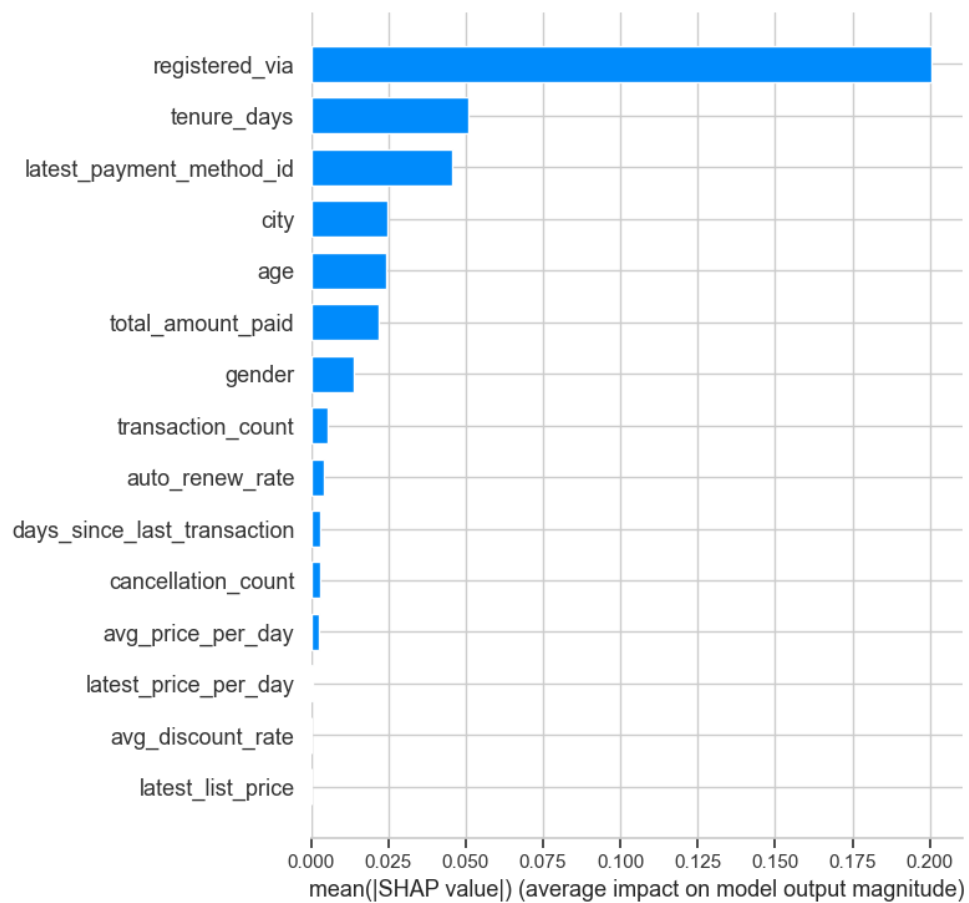


Figure 6: SHAP feature importance

Revenue Risk and Segment Prioritisation

The model output is converted into a business metric:

$$\text{risk-adjusted revenue exposure} = \text{churn probability} \times \text{expected revenue}$$

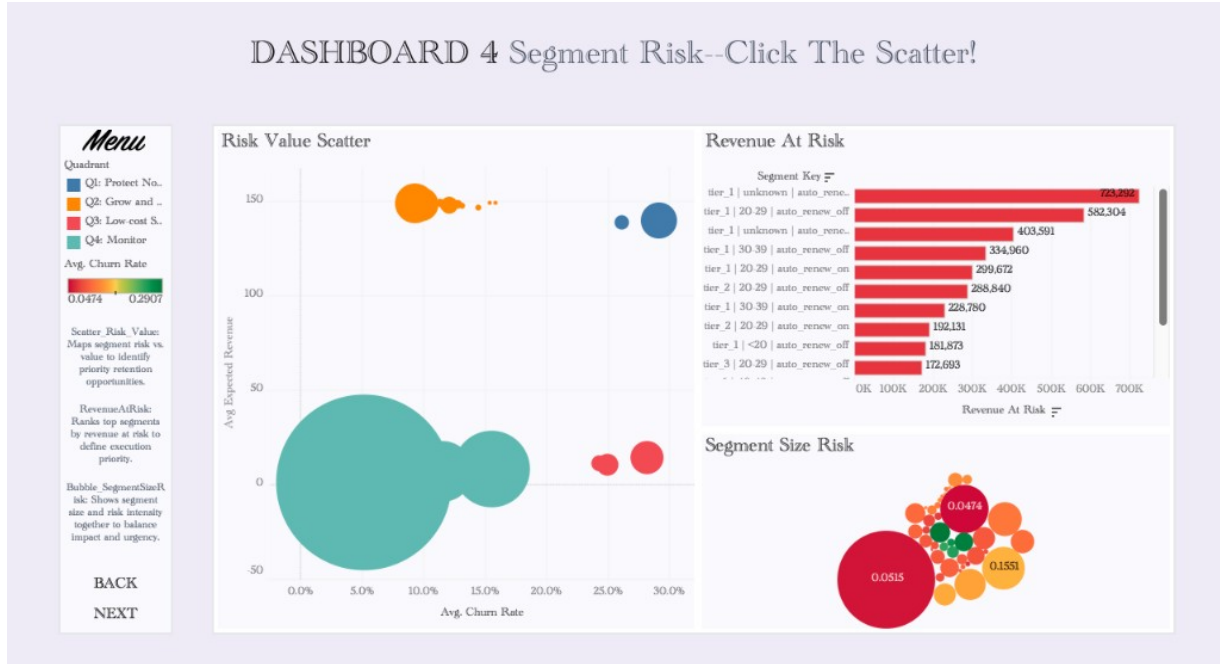


Figure 7: Segment risk dashboard

Risk band	Users	Average churn probability	Expected revenue	Risk-adjusted exposure	Recommended action
Critical	9,812	92.2%	3.04M	2.89M	Personalised offer + customer-service contact
High	4,438	70.5%	0.62M	0.43M	Targeted renewal discount test
Medium	215,515	40.3%	1.20M	0.47M	Low-cost re-engagement push
Low	741,195	16.9%	13.96M	0.85M	No active paid intervention

The critical and high-risk groups are only 1.47% of scored users but account for 71.5% of risk-adjusted exposure.

Revenue Impact and ROI Sensitivity

Revenue at risk is not the same as campaign value. In a real business, a retention campaign should be judged by incremental value:

$\text{expected saved revenue} = \text{targeted expected revenue} \times \text{churn reduction}$

$\text{discount cost} = \text{discount rate} \times \text{expected revenue} \times \text{expected renewal probability after treatment}$

$\text{net revenue impact} = \text{expected saved revenue} - \text{discount cost}$

$\text{ROI} = \text{net revenue impact} / \text{discount cost}$

The table below targets critical + high-risk users: 14,250 users and 3.67M price units of expected revenue.

Discount	Churn reduction	Expected saved revenue	Discount cost	Net revenue impact	ROI
10%	3pp	109,968	45,840	64,128	1.40
10%	6pp	219,936	56,837	163,099	2.87
10%	10pp	366,561	71,499	295,061	4.13
20%	3pp	109,968	91,680	18,288	0.20
20%	6pp	219,936	113,674	106,262	0.93
20%	10pp	366,561	142,999	223,562	1.56
30%	3pp	109,968	137,521	-27,552	-0.20
30%	6pp	219,936	170,511	49,425	0.29
30%	10pp	366,561	214,498	152,062	0.71

This analysis changes the recommendation. The model identifies who to consider, but ROI analysis decides what offer is sensible.

Retention Action Planner

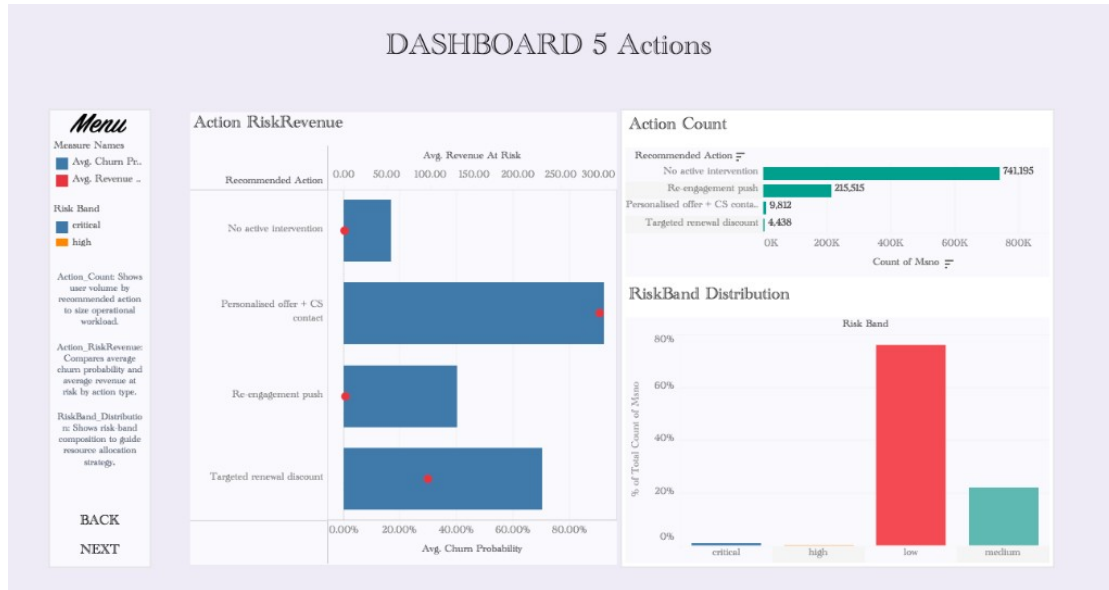


Figure 8: Retention action dashboard

The action dashboard translates model scores into operating rules. It shows that most users should not receive paid intervention, while high-risk groups should be handled with more targeted actions.

Recommended action logic:

- **Critical risk:** 9,812 users, 92.2% average churn probability, and 2.89M risk-adjusted exposure. Prioritise high-value users for personalised save offers and customer-service contact before expiry.
- **High risk:** 4,438 users, 70.5% average churn probability, and 0.43M risk-adjusted exposure. Run controlled renewal-incentive tests rather than broad campaign rollout.
- **Medium risk:** 215,515 users and 0.47M risk-adjusted exposure. Use low-cost re-engagement messages, product nudges, and playlist recommendations instead of discounts.
- **Low risk:** 741,195 users. Do not use paid incentives; monitor score movement and keep standard lifecycle communication.

Dashboard Suite

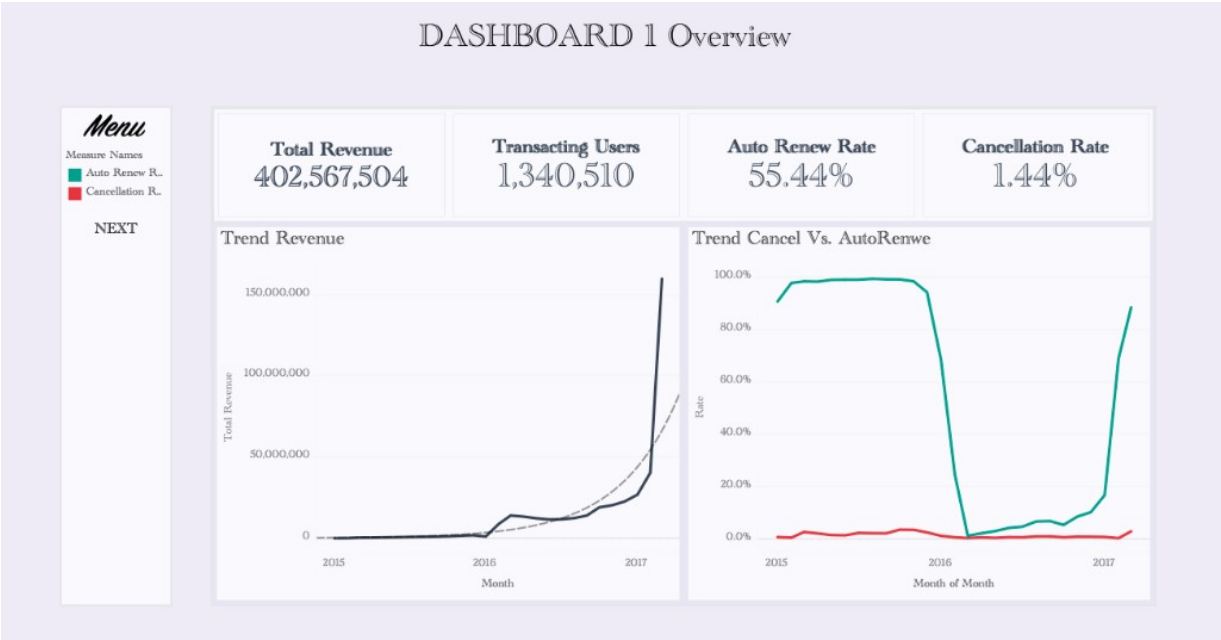


Figure 9: Executive overview dashboard

The overview dashboard summarises revenue, user volume, auto-renewal rate, cancellation rate, and monthly trend movement.

The churn drivers dashboard translates statistical outputs into business-readable evidence around auto-renewal, acquisition channel, discounting, and user profile patterns.

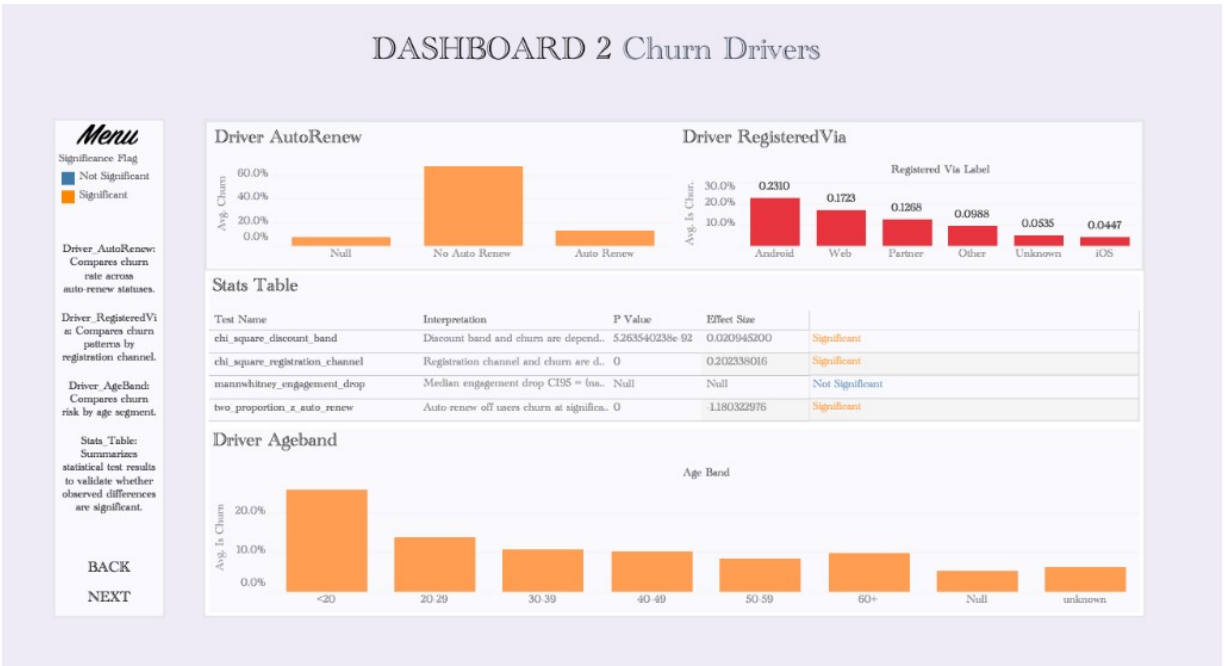


Figure 10: Churn drivers dashboard

The monitoring dashboard tracks AUC, top-decile recall, and PSI across cohorts. This shows whether model performance and score distributions remain stable after deployment.

Experiment Design

The recommended next step is a controlled A/B test, not immediate full rollout.

Design choice	Recommendation
Target users	High and critical risk users approaching expiry
Treatment	Renewal incentive through email and in-app banner
Control	Standard renewal reminder
Randomisation	User-level 50/50 split
Primary metric	30-day post-expiry retention
Guardrails	ARPU, complaint rate, auto-renew rate, net revenue after incentive cost
Decision rule	Roll out only if retention uplift and net revenue are both positive

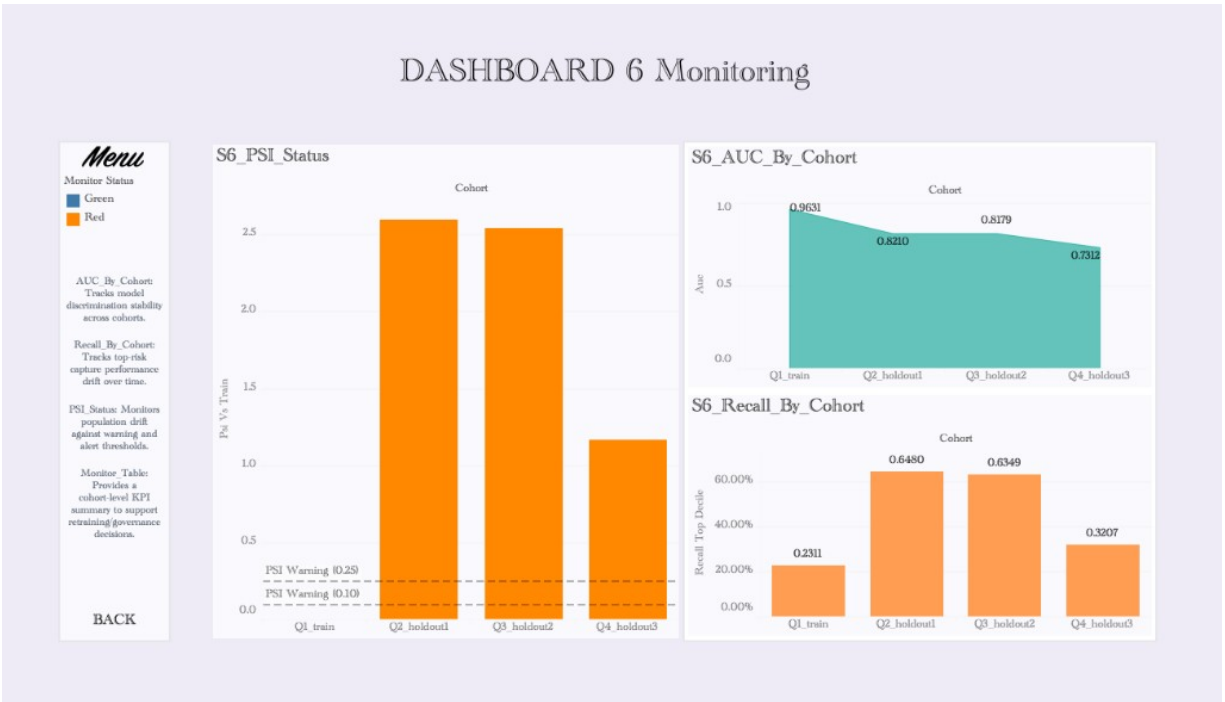


Figure 11: Monitoring dashboard

Final Recommendation

The recommended business strategy is to use the churn model as a **targeting and prioritisation layer**, not as an automatic discount engine. The model and dashboards show that risk is highly concentrated: critical and high-risk users represent only 1.47% of scored users but contribute 71.5% of risk-adjusted revenue exposure. This makes broad discounting inefficient.

1. Prioritise the critical-risk band first

The critical-risk band contains 9,812 users with 92.2% average churn probability and 2.89M price units of risk-adjusted exposure. These users should receive the highest-touch treatment, but only where expected revenue justifies the cost. Recommended actions are personalised save offers, customer-service contact, payment-continuity messaging, and expiry-date reminders.

2. Test incentives on the high-risk band

The high-risk band contains 4,438 users with 70.5% average churn probability and 0.43M price units of exposure. This group is suitable for a controlled renewal-incentive test. The model indicates high churn risk, but the ROI table shows that offer depth matters: a 10% discount with a 6pp churn reduction produces about 163,099 net revenue impact and 2.87 ROI, while a 30% discount with only 3pp reduction is negative at -27,552. Therefore, test smaller incentives first.

3. Use low-cost actions for medium-risk users

Medium-risk users are numerous, but their risk-adjusted exposure is much lower per user. The dashboard action view supports a lower-cost playbook: re-engagement emails, in-app prompts, playlist recommendations, and auto-renew reminders. This protects marketing budget while still nudging users whose scores may move upward.

4. Avoid paid intervention for low-risk users

Low-risk users account for the largest user count but should not receive paid discounts. The correct action is standard lifecycle communication and monitoring. This keeps retention spend focused on users where the model and revenue-risk dashboard show stronger business justification.

5. Validate with an A/B test before rollout

The final decision should be based on an experiment, not model scores alone. The proposed A/B test should target high and critical-risk users approaching expiry, use user-level randomisation, and evaluate 30-day post-expiry retention. Guardrails should include ARPU, complaint rate, auto-renew rate, and net revenue after incentive cost. Roll out only if retention uplift is statistically significant and net revenue remains positive.

Skills Demonstrated

- Analytics engineering with DuckDB and dbt.
- SQL data modelling from raw data to business marts.
- Statistical analysis and evidence-based churn driver validation.
- Machine learning model training, calibration, and lift-based evaluation.
- Revenue-risk analysis and ROI sensitivity.
- A/B test design and business decision rules.
- Tableau dashboard storytelling.
- Stakeholder-facing communication through a concise business report.